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| **Fiscal Year:** | FY 2008 | **Task Last Updated:** | FY 04/14/2008 |
| **PI Name:** | Fike, John R. Ph.D. | | |
| **Project Title:** | Inflammation in the brain after particulate irradiation predisposes the hippocampus to a heightened vulnerability after a secondary insult | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
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| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | (1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 94110 | **Congressional District:** | 8 |
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| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2004 Radiation Biology NNH04ZUU005N |
| **Start Date:** | 06/01/2005 | **End Date:** | 05/31/2010 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received NCE to 5/31/2010 per J. Dardano/JSC (3/2009)<br>NOTE: End date changed to 5/31/2009 per S. Krenek/JSC (8/07)<br>End date changed to 7/14/2009 per JSC info update (10/06) | | |
| **Key Personnel Changes/Previous PI:** | Add: Marta Andres-Mach, Ph.D. Post-doctoral Fellow | | |
| **COI Name (Institution):** | Obenaus, Andre ( Loma Linda University )<br>Noble, Linda ( University of California at San Francisco )<br>Gupta, Nalin ( University of California San Francisco )<br>Rosi, Susanna Ph.D. ( University of California San Francisco ) | | |
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| **Performance Goal Text:** | | | |

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| **Task Description:** | A unique feature of the space radiation environment is the presence of high-energy charged particles including protons and fully ionized atomic nuclei such as 56Fe. These particulate radiations may pose a significant hazard to space flight crews not only during the mission but also later, on Earth, when slow-developing adverse effects finally become apparent. In the CNS, exposure to such radiation may not only directly affect structure/function within the brain, but it may also make tissues more sensitive to a secondary insult such as trauma, stoke or degenerative disease. We have shown that photon and particulate irradiations induce a strong and persistent inflammatory response in the brain that is associated with altered hippocampal neurogenesis and impaired cognitive function. Based on these data we hypothesize that such inflammatory changes, while they may only lead to minor morphologic/functional changes by themselves, will predispose the brain, and in particular hippocampal neurogenesis, to a heightened vulnerability after a secondary insult. Secondly, we hypothesize that a pharmacologic reduction of radiation-induced inflammation will ameliorate the combined effects of radiation and secondary injury on hippocampal neurogenesis. While the nature of possible secondary insults is variable, we have identified traumatic brain injury (TBI) as the choice for the proposed studies. We have identified 4 specific aims to test our hypotheses: 1) determine if inflammatory changes induced by a low dose of proton irradiation predisposes the hippocampus to abnormal neurogenesis after secondary TBI; 2) determine if inflammatory changes induced by a low dose of 56Fe irradiation predisposes the hippocampus to abnormal neurogenesis after secondary TBI; 3) using knock-out mice lacking chemokine receptor 2 (CCR2), determine if a muted monocytic inflammatory response can reverse the influence of low dose proton and/or 56Fe irradiation on TBI-induced alterations in neurogenesis; and 4) determine if pharmacologic intervention after proton and 56Fe irradiation will reduce radiation-induced inflammation and lessen the effects of subsequent TBI. Mice will be irradiated with low radiation doses (0–2 Gy) and 30 days later will subjected to mild/moderate TBI. Neurogenesis and inflammation will be detected using cell specific immunohistochemistry and the effects of irradiation will be quantified using confocal microscopy and unbiased stereology. These studies will address a possible mechanism by which low-dose irradiation may sensitize the brain to secondary injury, and they will also involve a pharmacologic approach to minimize such effects. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Neurogenesis, or the generation of new neurons, continues throughout life in the hippocampal formation of all mammals, including humans. Neurogenesis is associated with hippocampal dependent cognitive functions, and reduced neurogenesis is linked to poorer cognitive function. Radiation alone and traumatic brain injury (TBI) alone both reduce neurogenesis and are correlated with hippocampal dependent cognitive impairments. Both of these insults occur in the context of normal life on earth, either via therapeutic use (irradiation) or as a result of accident or injury (TBI). Given that TBI is a relatively common type of injury, especially in young adults, it is conceivable that individuals receiving therapeutic irradiation for treatment of brain tumors, head and neck tumors, etc., could experience combined injury. Additionally, given the state of the world at this time, combined injury in a nuclear battlefield or in an uncontrolled radiation exposure such as a dirty bomb, is a possible outcome of a terrorist event. As a consequence, understanding the mechanisms associated with radiation, TBI alone or combined injury is essential for the development of suitable and effective countermeasures or treatments. |
| **Task Progress:** | Over the time period from the last progress report, we carried out 3 runs at BNL. The first and third runs involved 2-month old male wild type (WT) C57BL/6 and the second involved mice deficient in the gene for chemokine receptor 2 (CCR2). These latter animals have a deficient inflammatory response. Each treatment group (radiation alone, TBI alone, combined injury) consisted of 6 mice. Mice were anesthetized with isoflurane gas and given brain only doses of 56Fe irradiation (0, 0.5, 1, and 2 Gy). In runs 1 and 2, one month after irradiation mice were anesthetized using the agent Avertin and then received focal traumatic brain injury (TBI) one hemisphere of the brain. Sham-treated animals (controls) were subjected to the same surgical procedure but were not injured. Because of significant problems encountered in the first 2 studies (see below), in run 3 we used isoflurane gas anesthesia during both the irradiation and the induction of TBI. In all studies, 1 month following TBI mice received multiple injections of they thymidine analog bromodeoxyuridine (BrdU) to label newly born cells. Three weeks after the last BrdU injection, mice were anesthetized and perfused with paraformaldehyde. The brain was removed, processed, and sectioned for microscopic analysis. BrdU was used to label all newly born cells independent of cell type. We used cell specific antibodies to label newly born neurons, newly born astrocytes, and newly born activated microglia, the endogenous inflammatory cells of the brain. During the analysis of the tissues from the first BNL run we detected serious problems inasmuch as we saw 2-3 times the ‘normal’ mortality associated with TBI; normally TBI-induced mortality is about 3-5%. Additionally, in our control (sham treated) animals, the total numbers of newly born BrdU positive cells, independent of cell type, and newly born neurons were up to 8 times lower than what we have seen in previously reported studies. Without adequate control values, we were not confident about our measurements and were unable to determine how meaningful our data were. Unfortunately, we had completed our irradiations and TBI treatments in the CCR2 knock out (KO) mice (BNL Run # 2) before we had done enough analyses of the animals from Run # 1 to appreciate the potential problems described above. We found that the post-TBI mortality in our CCR2 study was over 50% in our un-irradiated controls and only slightly lower in the combined injury groups. Necropsy results in those animals showed necrosis of the gut with severe signs of inflammation. While the anesthetic Avertin has been reported to induce some changes after i.p. injection, in these KO animals that have a compromised immune response, the changes were overwhelming. Because we lost over 50% of the animals we did not analyze the brain tissues from these mice. Based on these data and other information, we concluded that there was a deleterious effect of the Avertin anesthetic in the context of our particular studies. This was a very unexpected finding, and while it may warrant further investigation in the future, it clearly has had a negative effect on our proposed experiments. The mechanism underlying this negative effect in not known.<br><br>Because of the complications cited above, we changed our protocol in Run # 3 such that both the irradiation and TBI procedures were carried out using isoflurane gas anesthesia. This resulted in no mortality during or after TBI, and cell numbers in sham treated controls were consistent with previous studies by us that did not specifically involve TBI. While the analysis of this most recent run is currently ongoing, we have accrued enough data from representative animals to convince ourselves that we will be able to obtain interpretable and meaningful data from this study. To date, our results suggest that radiation-induced changes, at least after the relatively low dose of 0.5 Gy, modulates (i.e. reduces) the extent of injury induced by a subsequent TBI. The precise mechanism(s) responsible are yet unclear but may involve neuroinflammation. The final analysis of our study will provide critical information as to the extent of this effect and if it depends upon radiation dose. This could have important ramifications regarding potential strategies or |

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| | approaches to the management of combined injury. Understanding the mechanism(s) responsible for the apparent 'preconditioning' effect could lead to potential pharmacologic interventions that could be used in the event of a post-irradiation traumatic event.<br><br>As a direct result of the anesthetic problems described above, we will be unable to fully complete all the studies as originally proposed, specifically Aim 4. However, in the next year we will be able to determine how proton irradiation affects subsequent TBI and if the effects are mediated through neuroinflammation. |
| **Bibliography Type:** | Description: (Last Updated: 04/11/2018) |
| **Abstracts for Journals and Proceedings** | Fike JR, Rola R, Potts MB, Tsuru K, Noble-Haeusslein L, Fishman K, Baure J, Rosi S, Milliken H, Obenaus A. "The effects of brain irradiation and subsequent traumatic brain injury on measures of forebrain neurogenesis." 18th Annual NASA Space Radiation Investigators' Workshop, Rohnert Park, CA, July 13-15, 2007.<br>Proceedings from 18th Annual NASA Space Radiation Investigators' Workshop, July 13-15, 2007, p. 152. , Jul-2007 |
| **Abstracts for Journals and Proceedings** | Fike JR, Rola R, Fishman K, Baure J, Obenaus A, Nelson G, Rosi S. "The effects of 56 Fe on measures of dentate neurogenesis." NASA Models of CNS Space Radiation Risks and Flight Interactions Workshop, Houston TX, October 30-31, 2007.<br>Proceedings from NASA Models of CNS Space Radiation Risks and Flight Interactions Workshop, October 30-31, 2007, p. 7. , Oct-2007 |